AERIS: Product Life cycle

Pre Phase A - Concept studies -
Mission Definition Review (MDR) → TRL 1-3

Phase A - Concept & Technology Development -
System Definition Review (SDR) & KDP-B → TRL 3-4

Phase B - Preliminary Design & Technology Completion -
Preliminary Design Review (PDR) → TRL 4-5

Phase C - Final Design & Fabrication -
Critical Design Review (CDR) & Production Readiness Review (PRR) → TRL 5-6

Phase D - System Assembly, Integration & Test -
Test Readiness Review (TRR) & System Acceptance Review (SAR) → TRL 7

Phase E - Operations & Sustainment -
Operational Readiness Review (ORR) & Periodic Mission Reviews → TRL 8-9

PLC Phase	FFBD Stage
Pre-Phase A	Stage 1 (MBU concept)
Phase A	Stages 1–2 (MBU + Filament prototyping)
Phase B	Stages 1–3 (through Gantry design)
Phase C	All stages (full fabrication)
Phase D	Full flow V&V
Phase E	FFBD Operational loop

TECHNICAL PROCESS

TRL 1-3

PRE-PHASE A
Concept Studies

4.1 Mission analysis & Requirements

- Define on-demand AM need: polymer, metallic, ceramic & ISRU parts in-situ (FR-1)
- Identify target environments: Manufacturing, Aerospace, Energy, Mining, Oil & Gas, Healthcare (ER-1.1)
- Establish mission objectives: reduce logistics mass, increase operational resilience, enable in-situ adaptation
- Document mission needs statement and reference missions; establish KPIs vs launching equivalent spare mass (C-3)
- Identify applicable standards: spaceflight safety, military standardisation (C-4)

4.2 Concept Trade Studies

- AERIS Desktop (226×226×366 mm, 2 kW peak / 1.5 kW continuous) vs AERIS Standard (914×914×1524 mm, 3 kW peak / 2.5 kW continuous) — OR-4
- WC screw extruder vs pellet-fed vs filament-fed approaches for polymer, metallic, ceramic printing
- Spectroscopy options trade study: SWIR (selected) vs Raman vs LIBS vs combination; gravimetric quality monitoring (FR-5)
- Ball milling (~1200 RPM) vs jet milling for feedstock particle sizing to 40–70 µm
- MBU configuration: dual centrifuge mixer + sieve + annealing chamber (200°C) + PEEK binder ≤2

4.3 Feasibility Assessment

- Ball-milling at ~1200 RPM for 3 min yields 40–70 µm rounded particles — confirmed via bench test
- PEEK binder <20 wt% of raw material; anneal at 200°C; slurry formation validated (FR-1, C-2)
- Power budget: 2–3 kW peak, 1.5–2.5 kW continuous — within host platform constraints (OR-4)
- Thermal control: ±2°C internal enclosure, . 40°C to +40°C operational range (ER-3)
- AERIS system can be built within cost, schedule and performance constraints — document evidence

4.4 Technology Gap Identification

- SWIR Spectroscopy Module (5.6) in-process QA — integration and calibration R&D
- 5-DOF autonomous post-processing arm with magnetic tool changer → prototype development path required
- ISRU feedstock sourcing and beneficiation for planetary surface operations (deferred to spaceflight variant)
- ML-reinforced defect detection (delamination, overhang, adhesion errors) → software prototype required
- Long-lead WC tooling procurement; anti-agglomeration nozzle and barrel qualification

5.1 Functional Requirement (Draft)

- Draft FR-1 through FR-6 set; operational requirements OR-1 through OR-5; constraints C-1 through C-5
- Success criteria SC-1 through SC-5 documented and tied to mission objectives
- Identify deferred spaceflight requirements: ER-1 (launch vibration/shock), ER-2 (microgravity), ER-4 (radiation tolerance)

Prepare for functional baseline release at SDR; initiate design-to specification tree

◆ MDR Gate
Feasible within cost, schedule & performance?

TRL 3-4

PHASE A
Concept & Technology Development

4.1 Refine Mission Requirements

- Finalize full requirement set: FR-1 to FR-6, OR-1 to OR-5, ER-1 to ER-6, IR-1 to IR-4, SC-1 to SC-5, C-1 to C-5
- Allocate requirements to subsystems: MBU (1.0), Filament Production (2.0), Print Engine Gantry (3.0), Post-Processing Arm (4.0)
- Sensing & Inspection Suite (5.0), Software & Data Management (6.0), Power & Thermal (7.0), Interface & Controls (8.0)
- Confirm deferred spaceflight items (ER-1, ER-2, ER-4) tracking plan established in risk register

4.2 System Architecture & WBS Definition

- Lock WBS 1.0 through 8.0; allocate to end items; confirm host platform interfaces (IR-1 mechanical, IR-2 electrical, IR-3 data)
- AERIS Desktop: 226×226×366 mm build volume, 2 kW peak / 1.5 kW continuous (OR-4)
- AERIS Standard: 914×914×1524 mm build volume, 3 kW peak / 2.5 kW continuous (OR-4)
- Define system assembly flow: MBU → heated filament maker → WC extruder → 3-axis Cartesian gantry → 5-DOF robotic arm
- Confirm SDOF robotic arm joint structure: Base (Yaw), Shoulder (Pitch), Elbow (Pitch), Wrist (Yaw) + magnetic tool changer

4.3Technology Development & Risk Mitigation

- WC nozzle hot-end bench trials: polymer (PEEK/PLA), metallic, ceramic print campaigns; nozzle wear characterisation
- Ball-mill particle sizing validation → 40–70 µm rounded particle morphology confirmed; process parameters locked
- PEEK binder anneal test at 200°C: slurry filament extrusion through heated funnel maker → wire filament output confirmed
- SWIR Spectroscopy Module (5.6) integration feasibility ML go/no-go gate software prototype (FR-5, FR-6)
- FFBD process flow validated: Material Beneficiation → Filament Preparation → Printing (Gantry) → Post-Processing

◆ MDR to SDR Gate
Architecture acceptable? Reqs complete? Buildable?

4.4 Concept Selction & Architecture Lock

- Tungsten carbide extruder (anti-agglomeration nozzle + barrel) + 3-axis Cartesian gantry selected — heritage FDM (C-2)
- 5-DOF robotic arm with magnetic tool changer platform; end-effector suite: gripper, saw, scraper tools
- HD camera systems selected for in-process inspection: HD Camera System in MBU (5.1), HD Camera System Gantry (5.2), HS Camera Array for Motion Capture (5.3), HS Camera for Post-Processing (5.4), HD Camera System for QC (5.10). ML reinforcement is provided by the ML Defect Detection Engine (6.2).
- Optional Raman/LIBS spectroscopy module (5.6) for material ID and in-process QC — deferred if cost-prohibitive
- Sealed vacuum chamber (optional, 3.6) and crucible assembly (3.7) for oxidation-sensitive materials

5.1 Functional Baseline Released (SDR)

- Functional baseline approved; design-to specification initiated for all 8 subsystems (1.0–8.0)
- End item identification: MBU, Filament Production Subsystem, Print Engine (Gantry), Post-Processing Unit, Sensing Suite
- System specification formally released at SDR; interface requirements (preliminary) IR-1 to IR-4 issued

5.2 Risk & Technology Mitigation Post-SDR

- Long-lead WC nozzle and barrel procurement initiated (C-2); stepper motor + planetary gearbox sourcing
- Raman camera qualification test plan drafted; arm safety protocol defined (ER-6: hot surfaces, moving parts)
- Deferred ER-1, ER-2, ER-4 tracked in risk register with design inputs for future spaceflight variant

TRL 4-5

PHASE B
Preliminary Design & Technology Completion

5.1 Preliminary Design - Subsystem Level

- Gantry (3.0): Al/composite frame, 3-axis Cartesian structure, precision linear rails, stepper/servo drivers with homing routines
- Thermal enclosure (3.5): ±2°C controlled, EMI shielding; vacuum chamber (3.6) optional for oxidation-sensitive materials (ER-3)
- Ceramic build plate (3.3): heated, auto-level array, removable substrate — FR-2
- MBU (1: Dual Centrifuge Mixer & Sieve (1.2, ≥20 kg capacity), Ball Mill Mechanism (1.3, ~1200 RPM), Thermal Annealing Chamber (1.4, 200°C), Binder Intake Module (1.5, PEEK ≤20%wt)
- Software (6.0): slicing & toolpath software (6.5), ML defect detection engine (6.2), arm control & path planning (6.3), QA decision engine (6.4)

5.2 Interface Definition & Control

- ICD: MBU → heated filament maker (2.2) → funnel transfer system (2.1) → WC extruder (2.3): mechanical, thermal & electrical
- Arm trigger ICD:Thermocouples (5.5) → Thermal trigger controller (5.9) → 5-DOF robotic arm (4.1) activation
- PDU (7.1): 2–3 kW peak budget; battery system (7.2); band heaters (7.3); cabling for power & encoder feedback (7.5)
- HMI (8.1): ruggedized interface screen; remote monitoring & command interface (1.8.2); data logging & downlink (6.5) — FR-3, OR-3
- Host platform interfaces: IR-1 mechanical mounts, IR-2 electrical (voltage/current limits), IR-3 telemetry/command

5.3 Engineering Development Tests

- Extruder print trials: layer adhesion, dimensional accuracy, nozzle wear rates — polymer, metallic, ceramic coupons
- Raman go/no-go gate calibration against certified reference standards; spectral drift characterisation
- Ball-milling process control validation: particle size distribution ±5 µm across 3-minute cycle
- Arm kinematics test: 5DOF reach envelope, magnetic tool changer exchange cycle, end-effector torque characterisation
- Thermal enclosure uniformity test: temperature distribution across build volume at SATP and -40°C / +40°C extremes (ER-3)

◆ PDR Gate
Design optimal? Meets req? Within cost/schedule?

5.4 Qualification Item Plans

- WC extruder assembly qualification plan (1.2.3): nozzle wear, anti-agglomeration performance, thermal cycling
- Robotic arm mechanism qualification plan (1.4.1): SDOF kinematics, torque limits, end-effector exchange, safety interlock
- Sensing suite qualification plan: HD Camera System in MBU (5.1), HD Camera System Gantry (5.2), HS Camera Array for Motion Capture (5.3), Thermocouples (5.5)
- Power & thermal management qualification: Power Distribution Unit (7.1), Battery System & Energy Storage (7.2), Band Heaters & Controllers (7.3)

5.5 Reliability & FMEA

- Critical failure modes: nozzle clog, arm collision, spectral drift, layer delamination, thermal runaway, slurry blockage
- Redundancy strategy: dual HD camera systems (5.1, 5.2, 5.4, 5.10)", backup end-effector tools, thermal cutoff, emergency stop (ER-6)
- FMEA analysis drives design changes to be resolved before CDR entry; failure mode registry maintained
- Reliability Program Plan: MTBF targets per subsystem; safety monitoring for crewed environment compliance (SC-3)

TRL 5-6

PHASE C
Final Design & Fabrication

4.1 Detailed Design - All Subsystems

- Tungsten Carbide Extruder Assembly (2.3): Extruder Drive (Stepper + Gearbox) (2.4), Band Heaters & Controllers (7.3), extruder barrel, anti-agglomeration nozzle, screw feed mechanism
- Print Head Assembly (3.2): nozzle + barrel; Ceramic Build Plate & Bed Levelling (3.3); Motion Control & Drive System (3.4)
- Slicing & Toolpath Software (6.5): CAD/STL → optimised G-code pipeline; CAD Repository & Material Profiles (6.6)
- Arm Control & Path Planning SW (6.3) — 5-DOF IK solver, collision avoidance, auto-trajectory planning; Arm Structural Base & Mounts (4.4)
- MBU drying & blending: Thermal Annealing Chamber (1.4) and Slurry Mixing Assembly (1.6) with mass flow meters; Feedstock Storage & Dispensing Module (1.8)
- Dimensional Inspection Sensor (5.7): structured-light/laser profilometer; Force / Torque Sensor (5.8)

4.2 Design Analysis

- Thermal FEA: enclosure heat loads, nozzle hot-end temperatures (up to print temp), build plate uniformity across volume
- Structural loads analysis: gantry frame under print forces, arm dynamic loads, handling and transport loads
- EMI/EMC compliance assessment — enclosure shielding effectiveness; band heater interference characterisation
- Power flow analysis: Power Distribution Unit design for 3 kW peak (7.1), battery sizing (7.2), band heater zone control (7.3)
- Kinematics analysis: 5-DOF arm workspace coverage vs build plate; magnetic tool changer cycle

4.3 Engineering Development Tests

- Print coupon campaigns: polymer (PEEK/PLA), metallic, ceramic — dimensional inspection vs CAD tolerances
- Dimensional inspection sensor suite validation: laser profilometer vs CAD tolerances; accept/reject threshold calibration
- QA Decision Engine (6.4): SWIR (5.6) + dimensional inspection (5.7) data → accept/reject log
- ML Defect Detection Engine (6.2) training: overhang, delamination, adhesion/layering errors — HD Camera System Gantry calibration (5.2
- 20-minute post-process cycle validation: arm support removal → defect detection → correction → sealed bin drop (FR-5

◆ CDR Gate
Detail design verified vs all reqs? Approved for production?

4.4 Interface Control Finalization

- Final ICDs signed off across all mechanical (IR-1), electrical (IR-2), software (IR-3), and modular (IR-4) interfaces
- Electrical interface to host platform: defined voltage/current limits; cabling for power & encoder feedback (7.5)
- Data interface: command, telemetry, downlink architecture finalised; data logging & downlink interface (6.5, IR-3)
- HMI ICD: Ruggedized Interface Screen (8.1); Remote Monitoring & Command Interface (8.2); Video Training Package (8.4)

5.1 PRR - Production Readiness

- Manufacturing plan and quality control procedures confirmed ready to begin end-item production
- Integration & Assembly Plan finalised; tooling, fixtures and test equipment available
- Production Readiness Review (PRR): proceed with production of all AERIS end items

5.2 Fabricate Qualification Items

- WC extruder assembly qualification unit: nozzle + barrel + stepper + gearbox — built and tested
- Robotic arm qualification unit: full 5DOF kinematics, end-effector tooling suite (gripper, saw, scraper) — validated
- SWIR Spectroscopy Module (5.6) qualification unit: calibration reference standards verified; ML model baseline locked
- Thermal management qualification unit: band heaters + PDU + battery system — thermal cycling

TRL 7

PHASE D
System Assembly, Integration & Test

6.1 Production Readiness

- Manufacturing plan and quality control procedures confirmed ready; PRR passed
- Integration & Assembly Plan finalised; tooling and fixtures available; production team trained
- Production Readiness Review (PRR): proceed with end-item production — all AERIS subsystems (1.1–1.8)

6.2 Fabricate End Items

- MBU (1: Dual Centrifuge Mixer & Sieve ≥ 20 kg (1.2), Ball Mill Mechanism (1.3), Thermal Annealing Chamber 200°C (1.4), Binder Intake Module PEEK (1.5)
- Filament Production (2): Funnel Transfer System (2.1), Heated Filament Maker (2.2), Tungsten Carbide Extruder Assembly (2.3), Extruder Drive (Stepper + Gearbox) (2.4), Hopper Assembly (2.5)
- Print Engine (Gantry) (3): 3-Axis Cartesian Gantry Structure (3.1), Print Head Assembly (3.2), Ceramic Build Plate & Bed Levelling (3.3), Motion Control & Drive System (3.4), Print Enclosure & Thermal Control (3.5), Crucible Assembly (3.7)
- Post-Processing Unit (4): 5-DOF Robotic Arm Mechanism (4.1), End-Effector Tooling Suite (4.2 — gripper/saw/scraper), Magnetic Tool Changer Platform (4.3), Sealed Part Collection Bin
- Sensing & Inspection Suite (5): HD Camera System in MBU (5.1), HD Camera System Gantry (5.2), HS Camera Array for Motion Capture (5.3), Thermocouples (5.5), Thermal Trigger Controller (5.9)
- Software & Data Management (6): Slicing & Toolpath Software (6.1), ML Defect Detection Engine (6.2), Arm Control & Path Planning SW (6.3), QA Decision Engine (6.4), Data Logging & Downlink Interface (6.5)
- Power & Thermal Management (7): Power Distribution Unit (7.1), Battery System & Energy Storage (7.2), Band Heaters & Controllers (7.3), Cooling Fans & Heat Dissipation (7.4); Interface & Controls (8): Ruggedized Interface Screen (8.1), Remote Monitoring & Command Interface (8.2)

6.3 System Integration & V&V

- Full AERIS system integration: all subsystems (1–8) interfaced, mechanically assembled and electrically commissioned
- Raman/LIBS camera array + calibration reference standards integrated and calibrated (5.6, optional)
- Data logging, downlink interface and remote command commissioning tested (FR-6, IR-3, OR-3)
- Software integration: slicing → G-code → gantry → arm IK → QA engine → data log — end-to-end pipeline validated
- FFBD full process flow verification: Material Beneficiation → Filament Prep → Printing → Post-Processing (4 main stages)

◆ TRR → SAR Gate
Test ready? Acceptance criteria met? System accepted?

6.4 Acceptance Testing

- Part coupon acceptance: polymer (PEEK), metallic, ceramic prints verified vs dimensional spec — FR-1, FR-2, SC-1
- 20-minute post-process cycle: arm support removal + ML defect detection + correction loop — FR-5, OR-2
- ML-reinforced camera defect detection: overhang, delamination, layering errors — all categories validated (FR-5)
- Crew safety verification: hot surface guards, arm e-stop, emission controls, proximity sensors active (ER-6, SC-3)
- Remote monitoring & command acceptance test: downlink, telemetry, commanding via HMI and remote interface (OR-3, IR-3)
- Power budget acceptance: Desktop ≤2 kW peak / Standard ≤3 kW peak; battery backup capacity confirmed (OR-4)

5.2V&V Compliance & As-Built Documentation

- Verification matrix fully closed — all FR/OR/ER/IR/SC requirements verified and documented
- As-built baseline documented; final design disclosure complete; interface control documents signed as-built
- Operations and maintenance manual finalised (OR-1); user instruction manual (8.3) and video training package (8.4) complete
- Training materials delivered; non-expert operator training programme validated (OR-1, SC-3)

TRL 8-9

PHASE E
Operations & Sustainment

7.1 Deploy & Install

- Delivery to target environment: Manufacturing, Construction, Aerospace, Energy, Mining, Oil & Gas, Healthcare, etc. (ER-1.1)
- Non-expert operator training with User Instruction Manual (8.3) and Video Training Package (8.4)— OR-1
- System integration with host platform: mechanical (IR-1), electrical (IR-2), data interfaces (IR-3)
- Operational Readiness Review (ORR): confirm system ready for operational use in target environment

OPS Operational Monitoring

- Remote monitoring & commanding via Ruggedized Interface Screen (8.1) and Remote Monitoring & Command Interface (8.2) — OR-3, FR-6
- Periodic SWIR Spectroscopy Module calibration maintenance against certified reference targets (5.6, optional)
- ML Defect Detection Engine (6.2) updates and QA Decision Engine (6.4) refinement — cruciible reloading as operational data accumulates
- Crew-proximity safety monitoring active: thermal monitoring, force-limited joints, emergency stop — ER-6 continuous compliance
- Data logging archive maintained; downlink and telemetry reporting active (FR-6, IR-3)

SUS Sustainment & Incremental Upgrades

- Incremental capability upgrades aligned with programme schedules — C-5
- End-effector tooling suite (gripper, saw, scraper) maintenance and replacement as wear occurs (1.4.2)
- Software updates: Slicing & Toolpath Software (6.1), Arm Control & Path Planning SW (6.3), QA Decision Engine (6.4), ML Defect Detection Engine (6.2)
- Feedstock Storage & Dispensing Module (1.8) replenishment; Binder Intake Module (1.5) PEEK procurement
- Extended operation with minimal maintenance — SC-2 evidence accumulates over operational lifetime

ASS Mission Assessment

- Mission partner acceptance confirmation — SC-5 evidence package maintained and reported
- Operational data downlink archive reviewed; performance against all success criteria assessed (SC-1 through SC-5)
- Part production quality statistics; dimensional repeatability, defect rates, material categories produced (SC-1)
- Periodic mission review reports submitted; maintenance and calibration logs archived

LL Lessons Learned & Spaceflight Transition

- Lessons learned captured for spaceflight variant development — operational reliability data, failure modes, maintenance burden
- Deferred requirements: ER-1 (launch vibration/shock), ER-2 (microgravity/partial gravity), ER-4 (radiation tolerance) — design inputs for spaceflight Phase A
- Reliability and maintenance data feeds into spaceflight qualification approach; MTBF actuals vs predictions
- Decommission and disposal plan executed; final lessons learned archive compiled and distributed

Key Reviews	**MDR** Mission Definition Review	**MDR → SDR → KDP-B** System Definition Review · Functional Baseline Exit	**PDR** Preliminary Design Review · Design-to Baseline Exit	**CDR · PRR** Critical Design + Production Readiness Review	**TRR · SAR** Test Readiness + System Acceptance Review	**ORR · Periodic Reviews** Operational Readiness + mission reviews
Deliverables	**Analysis & Planning** ▯ Mission needs statement ▯ Draft FR/OR/ER/IR/SC/C set ▯ Technology Dev Plan (TDP) ▯ Risk Mgmt Plan (draft) ▯ LCC estimate vs spare mass ▯ WBS Level 1–2 ▯ SEMP & Eng Master Plan	**Functional Baseline (SDR)** ▯ System specification released ▯ Design-to specs initiated 1.0-8.0 ▯ End item identification doc ▯ Interface Reqs IR-1..4 (prelim) ▯ TDP finalized · WBS full leafs ▯ FMEA (prelim): nozzle, arm, Raman	**Design-To Baseline (PDR)** ▯ Design-to specs all subsystems ▯ Updated ICDs ▯ Qual item plans (extruder/arm) ▯ V&V and acceptance plans ▯ FMEA report · Reliability Plan ▯ EMI/EMC & Contamination Plans	**Build-To Baseline (CDR)** ▯ Build-to specs (full disclosure) ▯ All ICDs signed ▯ Mtrls & Processes Control Plan ▯ Qual units + test results ▯ Mfg Plan · QA Plan ▯ Thermal FEA / structural results	**As-Built Baseline (SAR)** ▯ As-built documentation ▯ Verification matrix closed ▯ Acceptance test data packages ▯ Ops & maintenance manual ▯ Training materials + video ▯ ILS package · transition plan	**As-Deployed Baseline (ORR)** ▯ Partner acceptance records SC-5 ▯ Ops data downlink archive FR-6 ▯ Mission review reports ▯ Lessons learned reports ▯ Spaceflight variant inputs (ER-1/2/4) ▯ Decommission & disposal plan
Requirements	**FR-1 · FR-2 · FR-3** **OR-1 · OR-4** **ER-3 (thermal)** **C-2 (heritage) · C-3 (cost)** ER-1 · ER-2 · ER-4 DEFERRED	**FR-1...6 · OR-1...5** **ER-3 · ER-5 · ER-6** **IR-1...4 (prelim)** **SC-4 (TRL 3–4) · C-1 · C-4** ER-1 · ER-2 · ER-4 DEFERRED	**FR-1...6 (design verified)** **OR-2 (min crew time)** **OR-3 (remote monitoring)** **ER-3 · ER-5 · C-1** **SC-4 (TRL 4–5)**	**FR-1...6 (CDR verified)** **IR-1...4 (ICDs signed)** **SC-1 (repeatable parts)** **SC-4 (TRL 5–6) · C-1**	**SC-1 · SC-2 · SC-3 · SC-4** **FR-1...6 (matrix closed)** **OR-1 (training delivered)** **ER-6 (crew safety verified)** **TRL 7 milestone achieved**	**SC-1 · SC-2 · SC-5 ops evidence** **OR-1 · OR-3 active** **FR-6 continuous logging** **C-5 incremental upgrades** **TRL 8–9 mission-proven**
Technical Management	6.1–6.8 Technical Management ▯ 6.1 Programme schedule & TPMs ▯ 6.2 Requirements management ▯ 6.3 Risk register: WC nozzle, Raman, arm ▯ 6.4 LCC vs spare mass · WBS L1-2 ▯ 6.5 Info Mgmt Plan · SOW draft ▯ 6.6 MDR package preparation ▯ 6.8 Safety plan · ILS plan	6.1–6.8 Technical Management ▯ 6.1 Programme schedule & TPMs ▯ 6.2 Requirements management ▯ 6.3 Risk register: WC nozzle, Raman, arm ▯ 6.4 LCC vs spare mass · WBS L1-2 ▯ 6.5 Info Mgmt Plan · SOW draft ▯ 6.6 MDR package preparation ▯ 6.8 Safety plan · ILS plan	6.1–6.8 ▯ 6.1 PDR execution ▯ 6.2 Design-to baseline released ▯ 6.3 FMEA corrective actions ▯ 6.5 EEE Parts Mgmt Plan ▯ 6.8 Contamination Control Plan	6.1–6.8 ▯ 6.1 CDR/PRR planning ▯ 6.2 Build-to baseline released ▯ 6.3 Mfg Plan · QA Plan ▯ 6.8 Design change summary · waivers	6.1–6.8 ▯ 6.1 TRR/SAR planning ▯ 6.2 Acceptance test management ▯ 6.3 As-built baseline approved ▯ 6.5 Ops & maintenance manual ▯ 6.8 Training delivery · lessons learned	6.1–6.8 ▯ 6.1 ORR/mission readiness ▯ 6.2 Partner reporting · SC-5 evidence ▯ 6.3 Upgrade programme (C-5) ▯ 6.5 Lessons learned reports ▯ 6.8 Decommission plan execution